

Linear Systems of Equations

I. Linear Equation: a first degree polynomial

• Standard Form: $a_1x_1 + a_2x_2 + \dots + a_nx_n = b$

• Matrix Form: $\begin{bmatrix} a_1 & a_2 & \dots & a_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = b$

II. Linear Systems of Equations: a collection of first degree polynomials

• Standard Form:
$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{cases}$$

• Matrix Form: $\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \longrightarrow A\vec{x} = \vec{b}$

III. Solving Linear Systems

• using Gaussian Elimination

1. build augmented matrix: $[A|b]$

2. reduce matrix to Row Echelon Form

• REF if first nonzero in row is right of first nonzero in row above

• Infinite solutions: less equations than variables

• replace unknown variables with random variables

• No solutions: row with all zero coefficients and nonzero answer

Rank of Matrix: # non-zero rows when in REF

LU Decomposition

• any matrix A can be decomposed into an Upper, Lower, & Permutation matrix

$$A = ULP$$

• Elementary Matrix: a matrix that is 1 elementary row operation away from an identity matrix

• Properties for:

$$E_{ij} = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & c \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}$$

1. $E_{ij}^{-1} = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1/c \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}$

2. $E_{21}E_{31}\dots E_{m1} \cdot E_{32}E_{42}\dots E_{m2}\dots E_{mm-1} =$

3. $E_{ij(1)} \cdot E_{ij(2)} \rightarrow c = c_{ij(1)} + c_{ij(2)}$

$$\begin{pmatrix} 1 & & & & \\ c_{21} & 1 & & & \\ c_{31} & c_{32} & 1 & & \\ \vdots & c_{42} & & \ddots & \\ \vdots & \vdots & & & \ddots \\ c_{m1} & c_{m2} & \dots & c_{mm-1} & 1 \end{pmatrix}$$

• To find LU Decomposition:

1. reduce matrix to echelon form without scaling or interchanging rows
 $\rightarrow$ this produces U
2. apply each operation from step 1 to an elementary matrix and then multiply their inverses together in the same order they were used
 $\rightarrow$ this produces L

• if A has LU decomposition of $A=LU$, then: $\text{Rank}(A) = \text{Rank}(U)$
 $\det(A) = \det(U)$

• $\det(U)$ = product of diagonal values

• solving $A\vec{x} = \vec{b}$ using LU decompositions

1. decompose A into L & U
2. solve $L\vec{y} = \vec{b}$ for $\vec{y}$
3. solve $U\vec{x} = \vec{y}$ for $\vec{x}$

Matrix Norm & Error Analysis

- the norm of a vector $\vec{x} \in \mathbb{R}^n$ is: $\|\vec{x}\| = \sqrt{x_1^2 + \dots + x_n^2}$
- the norm of a matrix A is: $\|A\| = \max_{\vec{x}=1} \|A\vec{x}\|$ & $\|A^{-1}\| = \left(\min_{\vec{x}=1} \|A\vec{x}\| \right)^{-1}$
- the norm of diagonal matrix D is: $\|D\| = \max_i \{|d_i|\}$; where $d_1, \dots, d_n$ are diagonal values

• Error Analysis & Condition Number

• $A\vec{x} = \vec{b} \rightarrow A(\vec{x} + \Delta\vec{x}) = \vec{b} + \Delta\vec{b}$; a system where a minor error $\Delta\vec{b}$ results in $\Delta\vec{x}$

• Relative Error: $\frac{\|\Delta\vec{x}\|}{\|\vec{x}\|} \leq \|A\| \cdot \|A^{-1}\| \cdot \frac{\|\Delta\vec{b}\|}{\|\vec{b}\|}$

• Condition Number $\rightarrow \text{cond}(A)$

- represents greatest of effect that $\Delta\vec{b}$ has on $\Delta\vec{x}$
- when $\det(A) = 0$, $\text{cond}(A) = \infty$

• $A\vec{x} = \vec{b} \rightarrow (A + \Delta A)(\vec{x} + \Delta\vec{x}) = \vec{b}$

• Relative Error: $\frac{\|\Delta\vec{x}\|}{\|\vec{x} + \Delta\vec{x}\|} \leq \text{cond}(A) \cdot \frac{\|\Delta A\|}{\|A\|}$

Interpolation

- given a set of data points: $(t_0, y_0), (t_1, y_1), \dots, (t_d, y_d)$;
find a function $f(t)$ such that:

$$f(t_k) = y_k \text{ for } k = 0, 1, \dots, d$$

- there are infinitely many possible *interpolants* ($f(t)$); two assumptions are *polynomial interpolation* and *cubic spline*

- Polynomial Interpolation: $p(t) = c_0 \Phi_0(t) + \dots + c_d \Phi_d(t)$

- Monomial Basis ($\mathbb{P}_d$)

- the set of monomials: $\Phi_0(t) = 1, \dots, \Phi_d(t) = t^d$
forms a basis of $\mathbb{P}_d$

- given $d+1$ points, polynomial interpolation with respect to monomial basis forms a set of equations defined by:

$$\begin{bmatrix} 1 & t_0 & \dots & t_0^d \\ 1 & t_1 & \dots & t_1^d \\ \vdots & \vdots & \ddots & \vdots \\ 1 & t_d & \dots & t_d^d \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_d \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_d \end{bmatrix} \rightarrow A\vec{c} = \vec{y}$$

Vandermonde Matrix

- Properties

- $\det(A) = \prod_{0 \leq i < j \leq d} (t_i - t_j)$

- there exists a unique polynomial of degree $\leq d$ such that:

$$p(t_k) = y_k \text{ for } k = 0, \dots, n, \text{ given data points } (t_0, y_0), \dots, (t_n, y_n)$$

- Cubic Spline Interpolation

- piecewise defined function formed by N cubic polynomials:

$$f(t) = \begin{cases} P_1(t), & t_0 \leq t \leq t_1 \\ \vdots \\ P_N(t), & t_{N-1} \leq t \leq t_N \end{cases}; \quad P_k(t) = a_k(t-t_{k-1})^3 + b_k(t-t_{k-1})^2 + c_k(t-t_{k-1}) + d_k$$

for $k = 1, \dots, N$; such that: $f(t_i) = y_i$ for $i = 0, 1, \dots, N$;
and $f(t), f'(t), f''(t)$ are continuous

- the resulting Coefficient matrix is: $C = \begin{pmatrix} a_1 & \dots & a_N \\ b_1 & \dots & b_N \\ c_1 & \dots & c_N \\ d_1 & \dots & d_N \end{pmatrix}$
- $P_1 \leftarrow \quad \rightarrow P_N$

• 4 Conditions to use due to 4 variables (a_k, b_k, c_k, d_k) being in $P_k(t)$:

1. $P_k(t_{k-1}) = \gamma_{k-1}$ for $k \in [1, N]$

$\rightarrow d_k = \gamma_{k-1}$

2. $P_k(t_k) = \gamma_k$ for $k \in [1, N]$

$\rightarrow a_k(t-t_{k-1})^3 + b_k(t-t_{k-1})^2 + c_k(t-t_{k-1}) + d_k = \gamma_k$

3. $P'_k(t_k) = P'_{k+1}(t_k)$ for $k \in [1, N-1]$

$\rightarrow 3a_k(t-t_{k-1})^2 + 2b_k(t-t_{k-1}) + c_k = c_{k+1}$

4. $P''_k(t_k) = P''_{k+1}(t_k)$ for $k \in [1, N-1]$

$\rightarrow 6a_k(t-t_{k-1})^2 + 2b_k = 2b_{k+1}$

• 2 equations from natural cubic spline conditions:

1. $P'_1(t_0) = 2b_1 = 0$

2. $P''_N(t_N) = 6a_N(t_N - t_{N-1}) + 2b_N = 0$

• the coefficients C representing the cubic spline $p(t)$ are the solution of the system:

$$\begin{bmatrix} L_1^3 & L_1^2 & L_1 & 0 & 0 & 0 & \dots \\ 3L_1^2 & 2L_1 & 1 & 0 & 0 & 0 & \dots \\ 6L_1 & 2 & 0 & 0 & -2 & 0 & \dots \\ 0 & 0 & 0 & L_2^3 & L_2^2 & L_2 & 0 \\ 0 & 0 & 0 & 3L_2^2 & 2L_2 & 1 & 0 & -1 \\ 0 & 0 & 0 & 6L_2 & 2 & 0 & 0 & -2 \\ \vdots & \vdots & \vdots & & & & & \\ 0 & 0 & 0 & \dots & L_N^3 & L_N^2 & L_N & 0 \\ 0 & 2 & 0 & \dots & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \dots & 6L_N & 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ b_1 \\ c_1 \\ a_2 \\ b_2 \\ c_2 \\ \vdots \\ a_n \\ b_n \\ c_n \end{bmatrix} = \begin{bmatrix} \gamma_1 - \gamma_0 \\ 0 \\ 0 \\ \gamma_2 - \gamma_1 \\ 0 \\ 0 \\ \vdots \\ \gamma_N - \gamma_{N-1} \\ 0 \\ 0 \end{bmatrix} ; \text{ where } L_k = t_k - t_{k-1} \text{ and } d_k = \gamma_{k-1} \text{ for } k \in [1, N]$$

Orthogonality

• Subspaces: $U \subseteq \mathbb{R}^n$ is a **subspace** if:

1. U contains the zero vector
2. $\vec{u}_1 + \vec{u}_2 \in U$ for all $\vec{u}_1, \vec{u}_2 \in U$; closed under addition
3. $C\vec{u} \in U$ for all $C \in \mathbb{R}, \vec{u} \in U$

• Zero Subspace & $\mathbb{R}^n$ space are subspaces of $\mathbb{R}^n$

• Linear Independence and Span

• **span** of vectors $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_m \in \mathbb{R}^n$ is set of all linear combinations:

$$\begin{aligned} \text{span} \{ \vec{u}_1, \vec{u}_2, \dots, \vec{u}_m \} &= \{ c_1 \vec{u}_1 + c_2 \vec{u}_2 + \dots + c_m \vec{u}_m \in \mathbb{R}^n; c_1, c_2, \dots, c_m \in \mathbb{R} \} \\ &= \left(\vec{u}_1 \vec{u}_2 \dots \vec{u}_m \right) \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{pmatrix} \end{aligned}$$

• set of vectors $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_m \in \mathbb{R}^n$ are **linearly independent** if:

$$c_1 \vec{u}_1 + c_2 \vec{u}_2 + \dots + c_m \vec{u}_m = \vec{0} \text{ only when: } c_1 = c_2 = \dots = c_m = 0$$

$$\left(\vec{u}_1 \vec{u}_2 \dots \vec{u}_m \right) \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}; \text{ only trivial solution exists}$$

• Basis and Dimension

• set of vectors $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_m \in \mathbb{R}^n$ is basis of subspace $U \subseteq \mathbb{R}^n$ if:

1. $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_m$ are linearly independent

2. $\text{span} \{ \vec{u}_1, \vec{u}_2, \dots, \vec{u}_m \} = U$

• set of linearly independent vectors from U can form a **basis**

• # of vectors in basis is the **dimension**

• dimension also equals the rank of REF(U): $\dim(U) = \text{Rank}(\text{REF}(U))$

- Null Space: contains all solutions for $A\vec{x} = \vec{0}$

- $N(A) = \{\vec{x} \in \mathbb{R}^n : A\vec{x} = \vec{0}\}$

- with $A = LU$: $N(A) = N(U)$

- Range/Column Space of a Matrix

- $R(A) = \{\vec{y} \in \mathbb{R}^m : A\vec{x} = \vec{y} \text{ for some } \vec{x} \in \mathbb{R}^n\} = \{A\vec{x} \in \mathbb{R}^m : \vec{x} \in \mathbb{R}^n\}$

$$\longrightarrow R(A) = \text{span}\{\underbrace{\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n}_{\text{columns of } A}\}$$

- Inner Product (Dot Product):

$$\langle \vec{x}, \vec{y} \rangle = \vec{x} \cdot \vec{y} = \sum_{k=1}^n x_k y_k$$

- for complex vector: $\langle \vec{x}, \vec{y} \rangle = \sum_{k=1}^n x_k \overline{y_k}$ conjugate

- if A is $m \times n$ matrix, $\vec{x} \in \mathbb{R}^n$, $\vec{y} \in \mathbb{R}^m$, then:

$$\langle A\vec{x}, \vec{y} \rangle = \langle \vec{x}, A^T \vec{y} \rangle$$

- Orthogonality

- a set of vectors $\langle \vec{x}_1, \dots, \vec{x}_m \rangle \subseteq \mathbb{R}^n$ is an orthogonal set if:

$$\langle \vec{x}_i, \vec{x}_j \rangle = 0 \text{ for } i \neq j$$

- if such vectors are unit vectors ($\|\vec{x}_i\| = 1$ for all i), then they are an orthonormal set
- a set of orthogonal vectors is linearly independent (not necessarily vice-versa)

- Subspaces

- let $\{\vec{u}_1, \dots, \vec{u}_m\}$ and $\{\vec{v}_1, \dots, \vec{v}_n\}$ be bases of U_1 and U_2 respectively;

$$\longrightarrow U_1 \perp U_2 \text{ if } \langle \vec{u}_i, \vec{v}_i \rangle = 0 \text{ for all } i \in [1, m]$$

- Complement

- given $U \subseteq \mathbb{R}^n$, the orthogonal complement is the set:

$$U^\perp = \{\vec{x} \in \mathbb{R}^n : \langle \vec{x}, \vec{u} \rangle = 0 \text{ for all } \vec{u} \in U\}$$

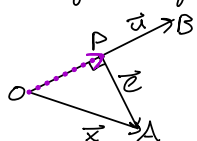
- $\dim(U) + \dim(U^\perp) = n$; ↖ $U \subseteq \mathbb{R}^n$

• Orthogonality Relations of Fundamental Subspaces

• given A is an $m \times n$ matrix, the four fundamental subspaces of A are:

- orthogonal ($^\perp$) subspaces
1. $N(A) = \{\vec{x} \in \mathbb{R}^n : A\vec{x} = \vec{0}\} \rightarrow$ null space of A
 2. $R(A) = \{\vec{y} \in \mathbb{R}^m : A\vec{x} = \vec{y} \text{ with } \vec{x} \in \mathbb{R}^n\} \rightarrow$ range / column space of A
 3. $N(A^T) = \{\vec{y} \in \mathbb{R}^m : A^T \vec{y} = \vec{0}\} \rightarrow$ null space of A^T
 4. $R(A^T) = \{\vec{x} \in \mathbb{R}^n : A^T \vec{y} = \vec{x} \text{ with } \vec{y} \in \mathbb{R}^m\} \rightarrow$ column space of A^T , row space of A

• Orthogonal Projections



find point P on $\vec{u}$ closest to $\vec{x}$

$$\vec{e} = \vec{x} - \text{proj}_{\vec{u}} \vec{x}$$

$$\text{proj}_{\vec{u}} \vec{x} = \frac{\vec{u} \cdot \vec{x}}{\vec{u} \cdot \vec{u}} \cdot \vec{u} = \boxed{\frac{1}{\|\vec{u}\|^2} \vec{u} \vec{u}^T} \cdot \vec{x}$$

red arrow $P \rightarrow$ projection matrix

• Properties:

1. $P^2 = P$
2. $P^T = P$
3. $\text{rank}(P) = 1$

• Vector Orthogonal to Subspace

- use *Gram-Schmidt algorithm* to find orthogonal basis:
- let $\{\vec{u}_1, \dots, \vec{u}_m\}$ be a basis of $U \subseteq \mathbb{R}^n$:

$$\begin{cases} \vec{v}_1 = \vec{u}_1 \\ \vec{v}_2 = \vec{u}_2 - \text{proj}_{\vec{v}_1} \vec{u}_2 \\ \vdots \\ \vec{v}_m = \vec{u}_m - \text{proj}_{\vec{v}_1} \vec{u}_m - \text{proj}_{\vec{v}_2} \vec{u}_m - \dots - \text{proj}_{\vec{v}_{m-1}} \vec{u}_m \end{cases}$$

$\rightarrow \{\vec{v}_1, \dots, \vec{v}_m\}$ is *orthogonal basis* of U ; all vectors are orthogonal: $\langle \vec{v}_i, \vec{v}_j \rangle = 0$

$\hookrightarrow \left\{ \frac{\vec{v}_1}{\|\vec{v}_1\|}, \dots, \frac{\vec{v}_m}{\|\vec{v}_m\|} \right\}$ is *orthonormal basis* of U

- Vector Projection onto Subspace

- Let $\{\vec{u}_1, \dots, \vec{u}_m\}$ be an *orthogonal basis* of $U \subseteq \mathbb{R}^n$;
projection of $\vec{x}$ onto U is:

$$\rho g u^x = \rho \bar{x}, \text{ where } \rho = \frac{1}{\bar{u}_1^T \bar{u}_1} \bar{u}_1 \bar{u}_1^T + \dots + \frac{1}{\bar{u}_m^T \bar{u}_m} \bar{u}_m \bar{u}_m^T$$

- let p represent the projection matrix onto U ; $I-p$ is the projection matrix onto $U^\perp$

- the $\text{proj}_U \vec{x}$ is the *closest vector* in U to $\vec{x}$:

$$\|\vec{x} - p_{ju}\vec{x}\| \leq \|\vec{x} - \vec{u}\| \text{ for all } \vec{u} \in U, \vec{u} \neq p_{ju}\vec{x}$$

- $$\cdot \|\vec{x} - \text{proj}_U \vec{x}\| = \text{proj}_U \vec{x} \rightarrow \text{shortest distance from } \vec{x} \text{ to } U$$

— RR Decomposition

- decompose a matrix into: $A = QR$
orthogonal $\leftarrow$ $\rightarrow$ upper triangle

- **Orthogonal Matrices:** columns and rows are orthonormal vectors

- a matrix is orthogonal if: $A^T A = A A^T = I$

- if A is an orthogonal matrix, then:

1. columns of A are orthonormal
2. rows of A are orthonormal
3. $\|A\vec{x}\| = \|\vec{x}\|$ for all $\vec{x} \in \mathbb{R}^n$

- Gauß-Schmidt QR Decomposition

- let A be an $n \times m$ matrix with $\text{rank}(A) = m$, find orthonormal basis $\{\vec{w}_1, \dots, \vec{w}_m\}$ via Gram-Schmidt algorithm of $R(A)$, and project columns of A ($\vec{a}_1, \dots, \vec{a}_m$) onto basis:

$$A = \underbrace{Q, R}_{\text{Thin QR decomposition}} = \begin{bmatrix} \vec{w}_1 & \dots & \vec{w}_m \end{bmatrix} \begin{bmatrix} \langle \vec{w}_1, \vec{a}_1 \rangle & \langle \vec{w}_1, \vec{a}_2 \rangle & \dots & \langle \vec{w}_1, \vec{a}_n \rangle \\ \langle \vec{w}_2, \vec{a}_1 \rangle & \langle \vec{w}_2, \vec{a}_2 \rangle & \dots & \langle \vec{w}_2, \vec{a}_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \vec{w}_m, \vec{a}_1 \rangle & \langle \vec{w}_m, \vec{a}_2 \rangle & \dots & \langle \vec{w}_m, \vec{a}_n \rangle \end{bmatrix}$$

- extend basis to $\{\vec{w}_1, \dots, \vec{w}_m, \vec{w}_{m+1}, \dots, \vec{w}_n\}$ of $\mathbb{R}^n$ where $\{\vec{w}_{m+1}, \dots, \vec{w}_n\}$ is any orthonormal basis of $\mathcal{R}(A)^\perp = \mathcal{N}(A^T)$

$$Q_2 = [\vec{w}_{m+1}, \dots, \vec{w}_n] \text{ resulting in: } A = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix} \begin{bmatrix} R_1 \\ O \end{bmatrix}$$

- Projection using QR: $\text{proj}_{\text{col}(A)} \vec{x} = Q_1 Q_1^T \vec{x}$

Least Squares Approximation

- in the event that $A\vec{x} = \vec{b}$ has *one or no solution*, find best approximation:
 - find $\vec{x}$ that minimizes $\|A\vec{x} - \vec{b}\|$ for best approximation
- 2 methods for finding LS approximation of $A\vec{x} = \vec{b}$:

1. $A^T A \vec{x} = A^T \vec{b}$

2. $R_1 \vec{x} = \vec{c}_1$ where $A = [Q_1 \ Q_2] \begin{bmatrix} R_1 \\ 0 \end{bmatrix}$ and $Q^T \vec{b} = \begin{bmatrix} \vec{c}_1 \\ \vec{c}_2 \end{bmatrix}$

- $\|A\vec{x} - \vec{b}\|^2 = \|R_1 \vec{x} - \vec{c}_1\|^2 + \|\vec{c}_2\|^2$

Fitting Models to Data

- Least squares data fitting will compute coefficients for:

$$f(t, \vec{c}) = c_1 f_1(t) + \dots + c_n f_n(t) \text{ to fit:}$$

$$(t_1, y_1), \dots, (t_m, y_m)$$

- Least Squares Polynomial Regression: best method for finding best fit

- Sum of Squared Errors: $\sum_{i=1}^m (y_i - f(t_i))^2 = \|\vec{y} - A\vec{c}\|^2$ where:

$$\vec{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}, \quad A = \begin{pmatrix} f_1(t_1) & \dots & f_n(t_1) \\ \vdots & & \vdots \\ f_1(t_m) & \dots & f_n(t_m) \end{pmatrix}, \quad \vec{c} = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}; \text{ solve } A\vec{c} = \vec{y}$$

Diagonalisation

- matrix A ($n \times n$) is diagonalisable if A has *n linearly independent eigenvectors*

- finding Eigenvalue (λ) & Eigenvector ($\vec{v}$):

- $\det(A - \lambda I) = 0$; solve for λ

- $\vec{v}_i = A - \lambda_i I$

- Characteristic Polynomial of A : $C_A(x) = \det(A - xI) \rightarrow$ roots are *eigenvalues*

- takes the form: $C_A(x) = \pm (x - \lambda_1)^{\alpha_1} \dots (x - \lambda_k)^{\alpha_k}$ ↑ algebraic multiplicity

- Defective Eigenvalue (diagonalisable if NOT defective)
 - λ_i is defective if the algebraic multiplicity (α_i) is greater than the geometric multiplicity ($d_i = \dim(E_{\lambda}) \leftrightarrow \#$ linearly independent eigenvectors of λ_i)
- Eigenspace for λ : $E_{\lambda} = \{\vec{v} : A\vec{v} = \lambda\vec{v}\}$
- if matrix A is diagonalisable, then there exists matrix P and D such that:

$$A = PDP^{-1}$$
 - columns of P are eigenvectors ($\vec{v}$) of A
 - entries of D are eigenvalues (λ) of A
 - A , which is $n \times n$, must have n linearly independent vectors
 - useful application: $A^3 = PD^3P^{-1}$
- Spectral Theorem
 - Let A be real and symmetric, then:

$$A = PDP^T; \text{ in other words } A \text{ is orthogonally diagonalisable}$$

Symmetric Matrices

- matrix A has only real eigenvalues
- matrix A is always diagonalisable
- all eigenvectors of distinct eigenvalues (i.e. $\lambda_i \neq \lambda_j$) are orthogonal (i.e. $\langle \vec{v}_i, \vec{v}_j \rangle = 0$)

Single Value Decomposition ($P\Sigma Q^T$)

- Let $\sigma_1 \geq \sigma_2 \geq \dots \sigma_r > 0$ be the singular values of A , then there exist matrices P & Q such that:

$$A = P\Sigma Q^T; \quad \Sigma = \begin{pmatrix} \sigma_1 & & & 0 \\ & \ddots & & \\ & & \sigma_r & \\ 0 & & & 0 \end{pmatrix}_{m \times n}$$

$\hookrightarrow$ where $\sigma_i = \sqrt{\lambda_i}$ when eigenvalues are ordered as: $\lambda_1 \geq \lambda_2 \geq \dots \lambda_r > 0$ → of AA^T or A^TA

- AA^T & A^TA are orthogonally diagonalisable: $AA^T = PD_1P^T$, $A^TA = QD_2Q^T$
- eigenvalues of AA^T is also eigenvalues of A^TA
 - all eigenvalues are non-negative ($\lambda \geq 0$)
 - geometric (d_i) & algebraic (α_i) multiplicity of λ is same for AA^T & A^TA

• SVD Construction (1 or 2)

1. Let $\vec{q}_k = \frac{1}{\sigma_k} A^T \vec{p}_k$, $k \in [1, r]$, where $\vec{q}_i$ are orthonormal eigenvectors of $A^T A$

• extend to $k \in [1, n]$ using Gram-Schmidt to find Q :

$$Q = \{\vec{q}_1, \dots, \vec{q}_r, \vec{q}_{r+1}, \dots, \vec{q}_n\} \text{ such that: } \begin{aligned} A^T A \vec{q}_k &= \sigma_k^2 \vec{q}_k, & k \in [1, r] & \& \\ A^T A \vec{q}_k &= 0, & k \in [r+1, n] & \& \\ \|\vec{q}_i\| &= 1 \end{aligned}$$

2. Let $\vec{p}_k = \frac{1}{\sigma_k} A \vec{q}_k$, $k \in [1, r]$, where $\vec{p}_i$ are orthonormal eigenvectors of AA^T

• extend to $k \in [1, n]$ using Gram-Schmidt to find P :

$$P = \{\vec{p}_1, \dots, \vec{p}_r, \vec{p}_{r+1}, \dots, \vec{p}_n\} \text{ such that: } \begin{aligned} A A^T \vec{p}_k &= \sigma_k^2 \vec{p}_k, & k \in [1, r] & \& \\ A A^T \vec{p}_k &= 0, & k \in [r+1, n] & \& \\ \|\vec{p}_i\| &= 1 \end{aligned}$$

• Properties, given $A = P \Sigma Q^T$

$$\begin{aligned} \cdot \text{rank}(A) &= r \\ \cdot \|A\| &= \sigma_1 \\ \cdot \|A^{-1}\| &= \sigma_r^{-1} \\ \cdot \text{cond}(A) &= \sigma_1 / \sigma_r = \sqrt{\lambda_{\max} / \lambda_{\min}} \\ \cdot N(A) &= \text{span}\{\vec{q}_{r+1}, \dots, \vec{q}_n\} & N(A^T) &= \text{span}\{\vec{p}_1, \dots, \vec{p}_r\} \\ \cdot R(A) &= \text{span}\{\vec{p}_1, \dots, \vec{p}_r\} & R(A^T) &= \text{span}\{\vec{q}_{r+1}, \dots, \vec{q}_n\} \end{aligned}$$

• Principle Component Analysis

• Let $\vec{x}_1, \dots, \vec{x}_n \in \mathbb{R}^p$ be a set of data such that $\sum_k \vec{x}_k = \vec{0} \rightarrow$ average of points is origin

$$\begin{aligned} \cdot "n" &\text{ is \# of samples} \\ \cdot "p" &\text{ is \# of features} \end{aligned} \quad X = \begin{pmatrix} \vec{x}_1 \\ \vdots \\ \vec{x}_n \end{pmatrix}_{n \times p}$$

• Let $\vec{w}_1 \in \mathbb{R}^p$ represent the direction of the most variance; the first weight vector:

$$\sum_{i=1}^n \langle \vec{x}_i, \vec{w}_1 \rangle^2 = \|X \vec{w}_1\|^2$$

• more generally, the k^{th} weight vector of X maximizes $\|X \vec{w}_k\|^2$; where:

$$X_k = X - \sum_{j=1}^{k-1} X \vec{w}_j \vec{w}_j^T$$

• Principle Component of $\vec{x}_i$: $\langle \vec{x}_i, \vec{w}_k \rangle$

- the *right vectors* $\vec{w}_1, \dots, \vec{w}_p$ are the *right singular vectors* $\vec{q}_1, \dots, \vec{q}_p$ of X
- $\|X \vec{w}_k\|^2$ is maximised when $\vec{w}_k = \vec{q}_k$

Pseudoinverse

- let $A_{m \times n} = P \Sigma Q^T$, the pseudoinverse of A is $A^+ = Q \Sigma^+ P^T$ where:

$$\Sigma^+ = \begin{pmatrix} \sigma_1^{-1} & & 0 \\ & \ddots & \\ 0 & & \sigma_r^{-1} & & 0 \\ 0 & & 0 & & 0 \end{pmatrix}_{n \times m}, \quad P = \{\vec{p}_1, \dots, \vec{p}_m\}, \quad Q = \{\vec{q}_1, \dots, \vec{q}_n\}$$

- if A is invertible, then: $A^+ = A^{-1}$
- the least squares approximation of $A\vec{x} = \vec{b}$ is: $\vec{x} = A^+ \vec{b}$
- SVD Expansion

$$A = \sum_{i=1}^r \sigma_i \vec{p}_i \vec{q}_i^T, \quad A^{-1} = \sum_{i=1}^r \sigma_i^{-1} \vec{q}_i \vec{p}_i^T$$

- Truncated SVD Expansion: $A_k = \sum_{i=1}^k \sigma_i \vec{p}_i \vec{q}_i^T$

Computing Eigenvalues

- Power Method: approximate eigenvalues of matrix

- let A be matrix with n eigenvalues such that: $|\lambda_1| \geq \dots \geq |\lambda_n|$ Dominant Eigenvalue
- let $\vec{x}_0$ be any *linear combinations* of eigenvectors of A : $\vec{x}_0 = \sum_{i=1}^n c_i \vec{v}_i$; $c_1 \neq 0$ eigenvectors
- then compute $\vec{x}_1, \dots, \vec{x}_k$:

$$\begin{aligned} \vec{x}_1 &= A \vec{x}_0 = \sum_{i=1}^n c_i \lambda_i \vec{v}_i \\ &\vdots \\ \vec{x}_k &= A^k \vec{x}_0 = \sum_{i=1}^n c_i \lambda_i^k \vec{v}_i \end{aligned} \quad ; \quad \left(\frac{\lambda_i}{\lambda_1} \right)^k \rightarrow 0 \text{ for } i > 1 \text{ \& } k \rightarrow \infty$$

→ therefore:

$$\frac{1}{\lambda_k} \vec{x}_k = \sum_{i=1}^n c_i \left(\frac{\lambda_i}{\lambda_k} \right) \vec{v}_i \doteq c_1 \vec{v}_1 \text{ as } k \rightarrow \infty$$

- Dominant Eigenvalue: λ with *algebraic multiplicity* of 1, and *greater magnitude* than all other eigenvalues

- Rayleigh Quotient: $\lambda = \frac{\vec{v}^T A \vec{v}}{\vec{v}^T \vec{v}}$
 - using the power method, the sequence of Rayleigh products (λ_k) converges to the dominant eigenvalue
- Normalised Power Iteration: normalise each $\vec{x}_k$ to shrink number:

$$\vec{x}_{k+1} = \frac{A \vec{x}_k}{\|A \vec{x}_k\|_\infty} \rightarrow \text{Inf Norm: } \|\vec{u}\|_\infty = \max \{|u_i|; i=1,2,\dots,n\}$$
- Inverse Iteration: find dominant eigenvalue of A^{-1} $\rightarrow$ smallest eigenvalue of A

$$\vec{x}_{k+1} = \frac{A^{-1} \vec{x}_k}{\|A^{-1} \vec{x}_k\|_\infty} = \frac{\vec{y}_{k+1}}{\|\vec{y}_{k+1}\|_\infty} ; \vec{y}_{k+1} = A^{-1} \vec{x}_k$$

————— Discrete Fourier Transform

- $z = a + bi = r e^{i\theta} ; \theta = \tan^{-1}\left(\frac{b}{a}\right), r = \sqrt{a^2 + b^2}$
- $e^{i\theta} = \cos\theta + i \sin\theta$
- $z^{-1} = \frac{\bar{z}}{|z|^2}$ ← conjugate
- Complex Inner Product

$$\langle \vec{v}, \vec{u} \rangle = v_1 \bar{u}_1 + \dots + v_n \bar{u}_n$$

$$\langle \vec{v}, \vec{u} \rangle = \overline{\langle \vec{u}, \vec{v} \rangle}$$
- Hermitian Matrix: matrix that is equal to its own conjugate transpose:

$$A = \bar{A}^T$$
 - a hermitian matrix allows: $\langle \vec{v}, A \vec{u} \rangle = \langle A \vec{v}, \vec{u} \rangle$
- Unitary Matrix: matrix whose inverse is equal to its own conjugate transpose:

$$A^{-1} = \bar{A}^T$$
 - a unitary matrix allows: $\|A \vec{v}\| = \|\vec{v}\|$
- Conjugate Matrix ($\bar{A}$): matrix whose elements are replaced by their conjugates

• Roots of Unity

- complex number ω such that: $\omega^N = 1$

$$\omega_N = e^{\frac{2\pi i}{N}} \rightarrow \omega_N \text{ is an } N^{\text{th}} \text{ root of unity along with } 1, \omega_N^2, \dots, \omega_N^{N-1}$$

- $\overline{\omega_N} = \omega_N^{-1} = \omega_N^{N-1}$
- $\omega_N^k = \cos\left(\frac{2\pi k}{N}\right) + i \sin\left(\frac{2\pi k}{N}\right)$
- $\sum_{n=0}^{N-1} \omega_N^{nk} = 0; k \in \mathbb{Z}, 0 < k < N$

• Fourier Basis:

$$\{\vec{f}_0, \dots, \vec{f}_{N-1}\} \text{ of } \mathbb{C}^N \text{ where } \vec{f}_k = \begin{bmatrix} 1 \\ \omega_N^k \\ \omega_N^{2k} \\ \vdots \\ \omega_N^{(N-1)k} \end{bmatrix}; \omega_N = e^{\frac{2\pi i}{N}}$$

- Fourier basis is an orthogonal basis of $\mathbb{C}^N$: $\langle \vec{f}_k, \vec{f}_l \rangle = \begin{cases} N; & k=l \\ 0; & k \neq l \end{cases}, \|\vec{f}_k\| = \sqrt{N}$
- $\vec{f}_k = \vec{f}_{N-k}$

• Discrete Fourier Transform:

$$F_N = \begin{bmatrix} \vec{f}_0^T \\ \vec{f}_1^T \\ \vdots \\ \vec{f}_{N-1}^T \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ | & \omega_N & \omega_N^2 & \dots & \omega_N^{N-1} \\ | & \omega_N^2 & \omega_N^4 & \dots & \omega_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ | & \omega_N^{N-1} & \omega_N^{2(N-1)} & \dots & \omega_N^{(N-1)^2} \end{bmatrix}, \text{ DFT}(\vec{x}) = F_N \vec{x}$$

- $\vec{y}[k] = \vec{y}[N-k]; \vec{y} = \text{DFT}(\vec{x})$
- $\text{DFT}(\vec{f}_k) = N \vec{e}_k$ for $k = 0, \dots, N-1$ where $\vec{e}_k$ is k^{th} standard basis vector

• Inverse Discrete Fourier Transform:

$$\text{IDFT}(\vec{y}) = \frac{1}{N} F_N^T \vec{y}$$

• Sinusoids

$$\vec{x} = A \cos(2\pi k \vec{T} + \phi) \quad ; \quad \vec{T} = \begin{pmatrix} 0 \\ 1/N \\ \vdots \\ \frac{N-1}{N} \end{pmatrix}$$

$\downarrow$ Amplitude
 $\downarrow$ Frequency
 $\downarrow$ Phase Shift

• Properties

$$1. \quad \vec{f}_k = \cos(2\pi k \vec{T}) \pm i \sin(2\pi k \vec{T})$$

$$2. \quad \frac{1}{2}(\vec{f}_k + \vec{f}_k^*) = \cos(2\pi k \vec{T})$$

$$3. \quad \frac{1}{2i}(\vec{f}_k - \vec{f}_k^*) = \sin(2\pi k \vec{T})$$

$$\cdot \text{ DFT}(\vec{x}) = \frac{AN}{2} e^{i\phi} \cdot \vec{e}_k + \frac{AN}{2} e^{-i\phi} \vec{e}_{N-k}$$

$$\cdot k=0: \text{ DFT}(\vec{x}) = AN \cos(\phi) \vec{e}_0$$

• Stemplots

• for complex vector $\vec{y} \in \mathbb{C}^N$:

$|y[n]|$ vs. n is the magnitude stemplot

$\angle y[n]$ vs. n is the angle stemplot